Subject: Biology

System: Endocrine and Nervous Systems

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disorder characterized by motor neuron dysfunction and loss leading to muscle atrophy and eventual death. A hallmark of ALS is the dysregulated excitability of neurons, which can manifest as hyperexcitability early in disease progression or hypoexcitability later in disease progression. Research has shown that this altered excitability occurs due to membrane ion channel dysfunction in motor neurons. In addition, altered levels of GABA, an inhibitory neurotransmitter, and glutamate, an excitatory neurotransmitter, also contribute to altered neuron excitability. ALS was originally thought to be a disorder solely of the motor division of the peripheral nervous system; however, it has been found that multiple body systems are affected and symptoms not related to the motor division, such as cognitive impairment, commonly develop over time.

Researchers hypothesized a relationship between the number of synapses (ie, synaptic density) in certain brain regions (frontal and motor cortexes) and the development of cognitive impairment symptoms in individuals with ALS. To investigate this, they measured synaptic density in the frontal and motor cortexes of patients affected with ALS who also demonstrated cognitive impairment (ALSci). These densities were compared to those of patients with ALS who did not demonstrate cognitive impairment (ALSnci) and also to those of healthy individuals without ALS (Table 1).

Table 1 Frontal Cortex and Motor Cortex Synaptic Densities in ALSci, ALSnci, and Healthy Patients: Means and 95% Confidence Intervals

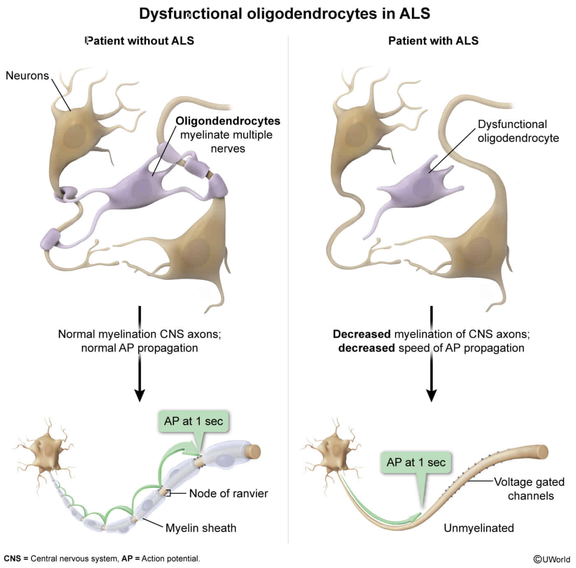
Patient condition	Frontal cortex synaptic density (synapses/mm^3)	Motor cortex synaptic density (synapses/mm^3)
ALSci	$6.75 \times 10^8 \pm 0.25 \times 10^8$	$7.25 \times 10^8 \pm 0.25 \times 10^8$
ALSnci	$7.50 \times 10^8 \pm 0.75 \times 10^8$	$7.10 \times 10^8 \pm 0.50 \times 10^8$
Healthy	$8.00 \times 10^8 \pm 0.25 \times 10^8$	$7.00 \times 10^8 \pm 0.50 \times 10^8$

Oligodendrocytes have been implicated in contributing to the disease phenotype in ALS. Cytoplasmic aggregates of mutant proteins within these cells prevent them from functioning properly. Given this fact, which of the following processes would be most affected in patients with ALS?

- A. Production of cerebrospinal fluid [2%]
- ✔ B. Myelination of central nervous system axons [88%]
- C. Phagocytosis of pathogens in the central nervous system [6%]
- D. Provision of nutrients to neurons [3%]

Correct
88% answered correctly

Explanation:



Cells of the nervous system include both **neurons** and **neuroglia** (glial cells). Neurons are responsible for sending electrical signals to other cells, and neuroglia provide support functions to neurons and the nervous system. Several types of neuroglia exist in the **central nervous system** (CNS):

- **Ependymal cells**: line compartments and produce cerebrospinal fluid **(Choice A)**.
- **Oligodendrocytes**: form myelin sheaths around axons to reduce ion leakage, decrease capacitance, and increase action potential propagation speed along the axon.
- **Microglia**: serve as immune cells which phagocytize pathogens, damaged cells, and waste materials **(Choice C)**.
- **Astrocytes**: contact blood vessels, regulating blood flow to coordinate synaptic activity and chemical changes. These cells are also important in maintaining extracellular fluid, ion, pH, and neurotransmitter homeostasis. In addition, astrocytes provide nutrients necessary for neuron function and are thought to play a role in neuronal development, maintenance, and communication with other glial cells **(Choice D)**.

Different types of neuroglia exist in the **peripheral nervous system** (PNS):

- **Schwann cells**: form myelin sheaths around axons to increase speed of conduction.
- **Satellite cells**: provide structural support and supply nutrients to neurons.

In this question, **oligodendrocytes** have been implicated in contributing to the disease phenotype of ALS. In this disease, cytoplasmic aggregates of mutant proteins prevent oligodendrocytes from functioning properly. Because **oligodendrocytes myelinate CNS axons**, this process would likely be most affected in patients with ALS.

Educational objective:

Glial cells exhibit functions that provide support to neurons in the nervous system. These cells include ependymal cells, oligodendrocytes, microglia, and astrocytes in the central nervous system and Schwann cells and satellite cells in the peripheral nervous system.

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